

## Ответы на домашнее задание №2

### Задание 1

Перевести из 10 в 16 систему 12345678, 1000000

The image shows handwritten calculations for converting decimal numbers to hexadecimal. For the first number, 1000000, the student uses a series of divisions by 16, with remainders 0, 4, 15 (F), 2, 4, 3, 2, 5, 1, 0, 0, 0, 0, 0, 0, 1. The final result is written as 1000000<sub>10</sub> = F4240<sub>16</sub>. For the second number, 12345678, the student uses a series of divisions by 16, with remainders 14 (E), 11 (B), 6, 1, 4, 1, 6, 0, 4, 7, 7, 1, 2, 3, 1, 2. The final result is written as 12345678<sub>10</sub> = BC614E<sub>16</sub>. Arrows indicate the order of remainders from bottom to top.

### Задание 2

Перевести из 16 в 10 систему 12345678, 1000000

$$\begin{aligned}
 & 1 \cdot 16^7 + 2 \cdot 16^6 + 3 \cdot 16^5 + 4 \cdot 16^4 + 5 \cdot 16^3 + 6 \cdot 16^2 + 7 \cdot 16^1 + 8 \cdot 16^0 \\
 &= 1 \cdot 268435456 + 2 \cdot 16777216 + 3 \cdot 1048576 + 4 \cdot 65536 + 5 \cdot 4096 + 6 \cdot 256 + 7 \cdot 16 + 8 \cdot 1 \\
 &= 268435456 + 33554432 + 3145728 + 262144 + 20480 + 1536 + 112 + 8 \\
 &= 305419896_{10}
 \end{aligned}$$

$$\begin{aligned}
 & 1 \cdot 16^6 + 0 \cdot 16^5 + 0 \cdot 16^4 + 0 \cdot 16^3 + 0 \cdot 16^2 + 0 \cdot 16^1 + 0 \cdot 16^0 \\
 &= 1 \cdot 16777216 + 0 \cdot 1048576 + 0 \cdot 65536 + 0 \cdot 4096 + 0 \cdot 256 + 0 \cdot 16 + 0 \cdot 1 \\
 &= 16777216 + 0 + 0 + 0 + 0 + 0 + 0 \\
 &= 16777216_{10}
 \end{aligned}$$

### Задание3

A – сгущенное молоко

B – мёд

C – хлеб

$A \&\& B \&\& !C$

### Задание4

Доказать тождества  $A \rightarrow B = !A \mid B$

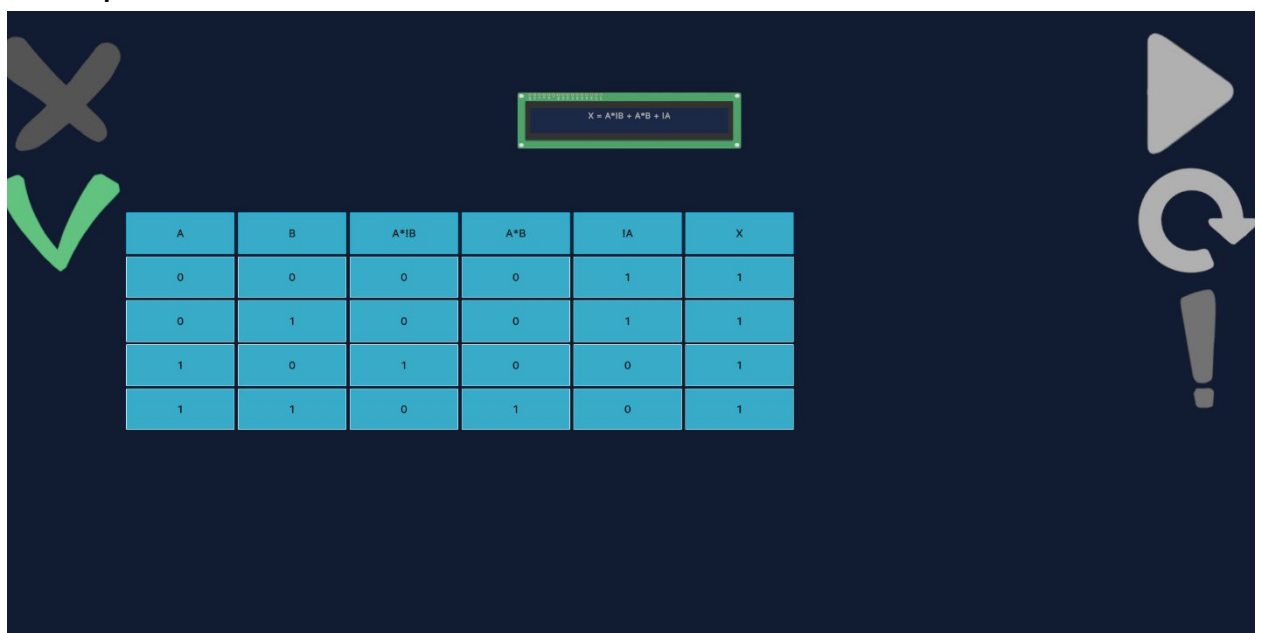
| A | B | !A | !A   B | $A \rightarrow B$ |
|---|---|----|--------|-------------------|
| 0 | 0 | 1  | 1      | 1                 |
| 0 | 1 | 1  | 1      | 1                 |
| 1 | 0 | 0  | 0      | 0                 |
| 1 | 1 | 0  | 1      | 1                 |

Доказать тождества  $A \leftrightarrow B = (A \&\& B) \mid \mid (!A \&\& !B)$

| A | B | $A \&\& B$ | $!A \&\& !B$ | $(A \&\& B) \mid \mid (!A \&\& !B)$ | $A \leftrightarrow B$ |
|---|---|------------|--------------|-------------------------------------|-----------------------|
| 0 | 0 | 0          | 1            | 1                                   | 1                     |
| 0 | 1 | 0          | 0            | 0                                   | 0                     |
| 1 | 0 | 0          | 0            | 0                                   | 0                     |
| 1 | 1 | 1          | 0            | 1                                   | 1                     |

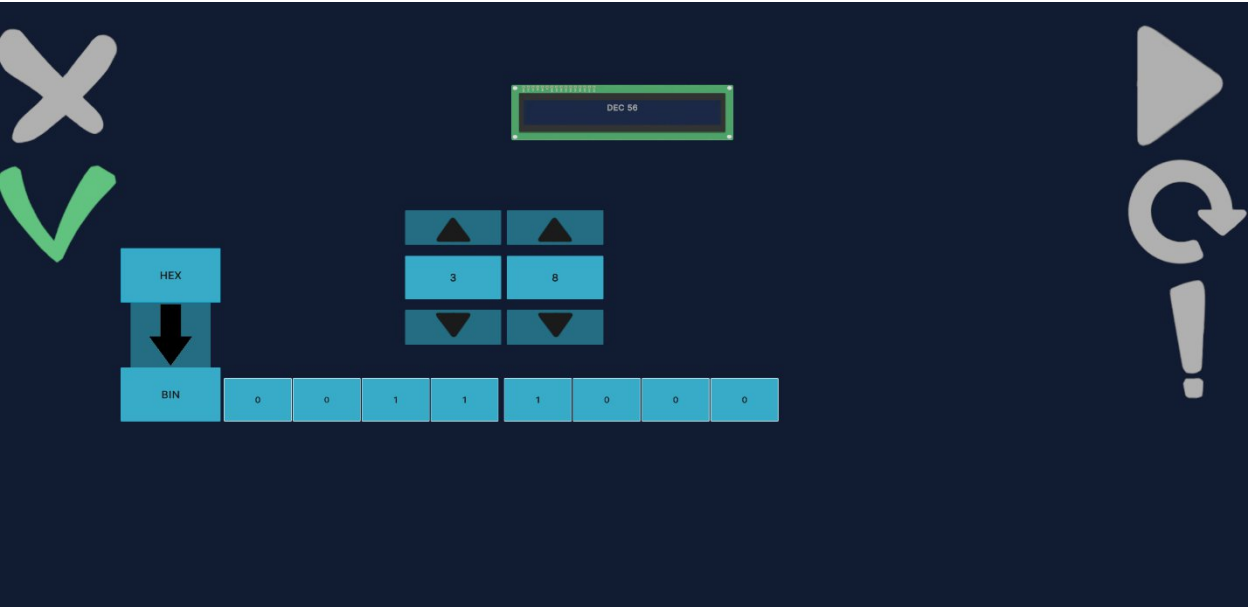
### Задание5

Таблица истинности

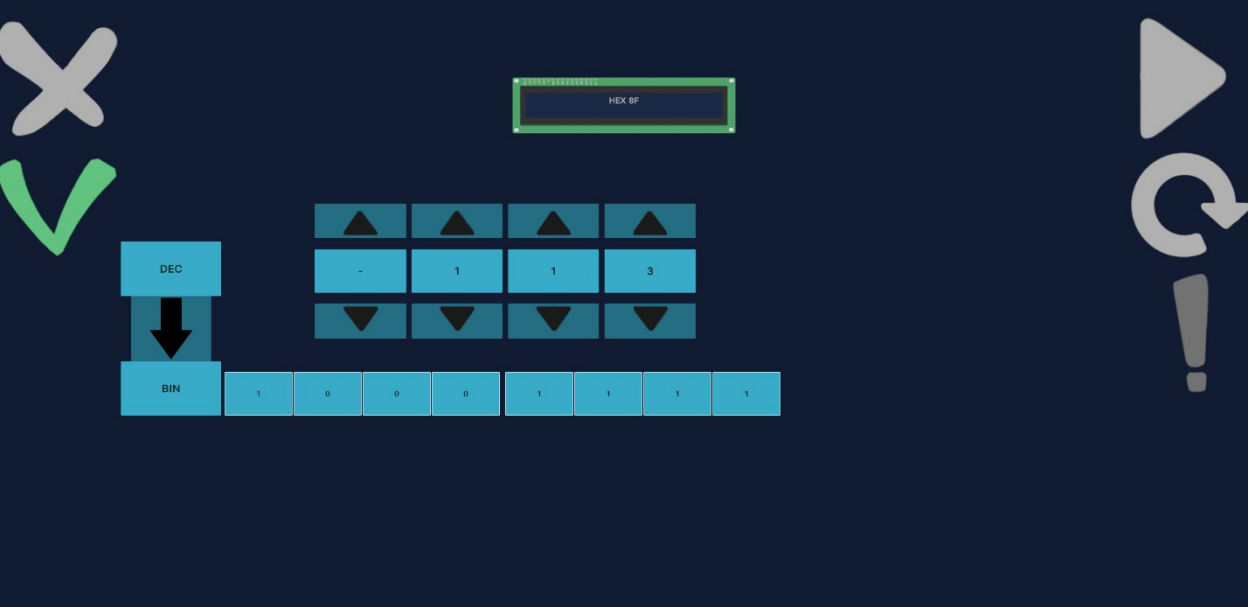


| A | B | $A \cdot !B$ | $A \cdot B$ | $!A$ | X |
|---|---|--------------|-------------|------|---|
| 0 | 0 | 0            | 0           | 1    | 1 |
| 0 | 1 | 0            | 0           | 1    | 1 |
| 1 | 0 | 1            | 0           | 0    | 1 |
| 1 | 1 | 0            | 1           | 0    | 1 |

Система счисления



Дополнительный код



# Задание 6

$$X = (B \rightarrow A) \cdot (\overline{A+B}) \cdot (A \rightarrow C)$$

$$X = (!B + A) \cdot (!A \cdot !B) \cdot (!A + C)$$

$$X = ((!B \cdot \overset{!A \cdot !B}{!A \cdot !B}) + \overset{A=0}{A \cdot !A \cdot !B}) \cdot (!A + C)$$

$$X = !A \cdot !B \cdot (!A + C)$$

$$X = \overset{!A \cdot !B}{!A \cdot !B} \cdot !A + !A \cdot !B \cdot C$$

$$X = !A \cdot !B + !A \cdot !B \cdot C$$

По закону поглощения упрощаем

$$X = !A \cdot !B$$